

Lab 9: Gravimetric Analysis of Chloride

Chemistry 150 - Winter 2026

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Lab Section	2026W CHEM-150-X01AB
Date Performed	March 23, 2026
Unknown Number	379

Data

Table 1. Raw measurements for both trials.

Measurement	Trial 1 (Dylan)	Trial 2 (David)
Mass of unknown (g)	0.4160	0.4170
Initial mass of filter paper (g)	0.4690	0.7280
Final mass of filter paper + AgCl (g)	1.4050	1.6630
Mass of AgCl (g)	0.9360	0.9350

Calculations

$$\% \text{Cl}^- = \frac{m_{\text{AgCl}} \times 35.45}{143.32 \times m_{\text{unknown}}} \times 100\%$$

Trial 1:

$$\% \text{Cl}^- = \frac{0.9360 \times 35.45}{143.32 \times 0.4160} \times 100\% = 55.65\%$$

Trial 2:

$$\% \text{Cl}^- = \frac{0.9350 \times 35.45}{143.32 \times 0.4170} \times 100\% = 55.46\%$$

Table 2. Summary of results.

	Trial 1	Trial 2
Mass of AgCl (g)	0.9360	0.9350
Moles of Cl ⁻ (mol)	0.006531	0.006524
Mass of Cl ⁻ (g)	0.2315	0.2313
% Cl ⁻	55.65%	55.46%
Average % Cl⁻	55.56%	

Conclusion

The chloride content of unknown 379 was found to be **55.56% Cl⁻** (average of two trials). The trials agreed to within 0.19 percentage points.